FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Benign Lesions of the Cervix

Tutorial | Week 4 · Hypertension, twins & the pelvic floor | Gynaecology | Dr Kaduyu

Most cervical complaints (discharge, post-coital bleeding, a "red cervix" or a lump at the os) are benign — ectropion, cervicitis, polyps, Nabothian cysts or fibroids — but every one of them is a diagnosis made only after cervical cancer has been actively excluded by looking, screening and, if anything looks suspicious, biopsying.

Must know before the ward round

1. The transformation zone (TZ) lies between the original and the new squamocolumnar junction; it is where metaplasia happens, where Nabothian cysts form and where almost all cervical cancers start.
2. Ectropion (ectopy) is columnar epithelium on the ectocervix — physiological with puberty, pregnancy and the combined pill; treat only if symptomatic AND screening is normal.
3. Mucopurulent endocervical discharge + a friable cervix that bleeds on touch = cervicitis (gonorrhoea/chlamydia) until proven otherwise; cervical motion tenderness means it has ascended (PID).
4. Uganda syndromic treatment of cervicitis — cefixime 400 mg PO stat (or ceftriaxone 1 g IM stat) + doxycycline 100 mg PO 12-hourly for 7 days; in pregnancy replace doxycycline with azithromycin 1 g PO stat. Always treat the partner(s).
5. Post-coital bleeding is cervical cancer until the cervix has been seen, screened and (if anything looks abnormal) biopsied — never treat it blindly with antibiotics or reassurance.
6. Cervical polyps are removed by avulsion (twisting) in clinic and ALWAYS sent for histology; in a post-menopausal woman also assess the endometrium.
7. Nabothian cysts are harmless mucus-retention cysts of the TZ — no treatment; their presence simply tells you where the TZ is.
8. A central cervical fibroid distorts the pelvis and pushes the ureters laterally — the classic source of ureteric injury at hysterectomy.
9. VIA uses 3–5% acetic acid read at 1 minute; an eligible VIA-positive lesion (SCJ fully seen, under 75% of the ectocervix, no canal extension, not suspicious, not pregnant) is ablated the same day — thermal ablation at ≥100 °C for 20–30 s or cryotherapy 3-5-3.
10. Outpatient PID — ceftriaxone 1 g IM stat + doxycycline 100 mg 12-hourly + metronidazole 400–500 mg 12-hourly, both for 14 days; always do a pregnancy test first (ectopic) and admit if no improvement in 72 h.
11. Pregnancy test before doxycycline, fluconazole or any procedure; bleeding after about 20 weeks means ultrasound to exclude placenta praevia before any speculum.
12. Failure of one full syndromic course means check adherence and partner treatment, then swab for NAAT/culture and sensitivities (Ugandan resistance to doxycycline, azithromycin and cefixime) — never just repeat the packet.

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1. Definition

The cervix is the most visible, most examined and most worried-about organ in the gynaecology clinic. In Uganda, where cervical cancer is the commonest cancer in women, any woman with **abnormal discharge, post-coital bleeding (PCB), inter-menstrual bleeding or an "abnormal-looking" cervix** makes both her and the clinician think of cancer. Most of the time the cause is benign: **ectropion, cervicitis, a polyp, Nabothian cysts, or a cervical fibroid**. These are easy to recognise once you understand the **anatomy of the transformation zone**, and easy to manage, provided you follow one rule:

> **RED FLAG**
> **Always exclude malignancy first.** A benign diagnosis is a diagnosis of exclusion: look at the cervix with a speculum, make sure she is up to date with screening (VIA or HPV test), and **biopsy anything that looks suspicious** (ulcer, friable mass, irregular growth, contact bleeding that does not settle after treating infection).

The second thing to master is **syndromic management of vaginal discharge** as practised in Uganda, because cervicitis caused by *Neisseria gonorrhoeae* and *Chlamydia trachomatis* is common, often silent, and is the root of much of Uganda's **PID, ectopic pregnancy and tubal infertility**.

Term	Definition
Ectocervix	Vaginal part of the cervix, covered by pink, smooth, **non-keratinised stratified squamous epithelium**
Endocervix	Canal from external to internal os, lined by a **single layer of mucus-secreting columnar epithelium** thrown into crypts ("glands")
Original squamocolumnar junction (SCJ)	Where squamous meets columnar epithelium at birth/childhood
New (functional) SCJ	Where the junction lies now, after metaplasia has replaced exposed columnar cells with squamous cells
Transformation zone (TZ)	Area between the original and new SCJ: the site of metaplasia and of almost all CIN and squamous cancer
Ectropion (ectopy / "erosion")	Columnar epithelium visible on the ectocervix, giving a red, velvety area around the os. "Erosion" is an old, misleading word: the epithelium is not eroded
Cervicitis	Inflammation of the cervix. **Acute**: mucopurulent discharge, oedema, friability, usually infective. **Chronic**: persistent low-grade inflammation, often histological, with discharge
Nabothian cyst	Mucus-retention cyst formed when squamous metaplasia covers the opening of an endocervical crypt
Cervical polyp	Pedunculated or sessile benign overgrowth of endocervical (commonest) or ectocervical tissue
Cervical fibroid (leiomyoma)	Fibroid arising in the cervix (about 1–2% of all fibroids); may be interstitial, sub-serous, or a pedunculated fibroid prolapsing through the os
Cervical stenosis	Narrowing of the canal, usually at the internal os, that may cause haematometra/pyometra
Cervical insufficiency ("incompetence")	Painless dilatation of the cervix in the second trimester without contractions, causing mid-trimester loss or preterm birth

2. Epidemiology

- **Ectropion** is found in a large proportion of adolescents, pregnant women and COC users: it is a normal variant, not a disease.
- **Nabothian cysts** are among the most common incidental findings on speculum examination and pelvic ultrasound.
- **Cervical polyps** occur in roughly 2–5% of women, commonest in multiparous women in their 40s and 50s. Malignant change is rare (well under 1–2%), but that is exactly why every polyp is sent to histology. Recurrence is about 6%.
- **Cervical fibroids** make up about 1–2% of all fibroids.
- **Cervicitis / STIs**: gonorrhoea and chlamydia are common in young sexually active Ugandans, and most infected women are **asymptomatic**. Untreated, 10–15% ascend to PID, which then drives tubal infertility and ectopic pregnancy: hence the very high secondary infertility in the region.
- **Antimicrobial resistance** is real: a 2023 study from a tertiary hospital in south-western Uganda found high resistance of vaginal isolates to **doxycycline, azithromycin and cefixime** (the very drugs used in syndromic packages), which is why follow-up and partner treatment matter so much.
- Most women with vaginal discharge in Uganda are managed at **HC II–IV** level using **syndromic management** (treatment by clinical syndrome without laboratory confirmation) because Gram stain, NAAT/GeneXpert for CT/NG and culture are rarely available at the point of care.

> **UGANDA CONTEXT**
> The Ministry of Health approach (Uganda Clinical Guidelines, "Abnormal vaginal discharge syndrome") is **syndromic**: take a history and risk assessment, do a speculum examination if possible, and treat for the likely group of organisms (cervical infection and/or vaginal infection), with partner notification, condoms and HIV testing: the "**4 Cs**": **Compliance, Counselling, Condoms, Contact tracing**.

Services by level of care (detail of tests in section 8 and of procedures in section 9) Verify *compiled from the source notes, not an MoH service list*

Level	What you can do
HC III	History, speculum, VIA (where trained), syndromic treatment, partner slips, HIV test, referral
HC IV	All the above + polypectomy, cryotherapy/thermal ablation, HPV sample collection (GeneXpert hubs), management of mild PID, admission for severe PID, basic ultrasound
General/regional referral hospital	Colposcopy, LEEP, punch biopsy + histology, cultures/NAAT, EUA, vaginal myomectomy, cerclage, hysterectomy for fibroids, drainage of TOA/pyometra
National referral / UCI	Gynae-oncology for suspected malignancy

3. Aetiology and Risk Factors

Lesion	Cause / driver	Risk factors
Ectropion	**Oestrogen** makes the cervix bulk up and evert	Puberty and adolescence, **pregnancy**, **combined oral contraceptive**
Acute cervicitis	*C. trachomatis*, *N. gonorrhoeae* (main); also *Mycoplasma genitalium*, trichomonas, herpes simplex; rarely schistosomiasis, TB	Positive **risk assessment**: partner with urethral discharge or genital ulcer; age under about 21–25; new partner in the last 3 months; more than one partner; no condoms; HIV

Lesion	Cause / driver	Risk factors
Chronic cervicitis	Long-standing low-grade inflammation	Childbirth lacerations, repeated infection
Condylomata	HPV 6/11	HIV
Nabothian cysts	Squamous metaplasia covering crypt openings	Physiological (TZ activity), chronic cervicitis
Endocervical polyp	Focal hyperplasia of endocervical glands and stroma, probably driven by **chronic inflammation and oestrogen**	Multiparity, age 40s–50s
Cervical fibroid	Smooth-muscle tumour like uterine fibroids	As for uterine fibroids
Cervical stenosis	Scarring or atrophy	LEEP/cone, cryotherapy, radiotherapy, cervical cancer, oestrogen deficiency after menopause
Cervical insufficiency	Weak or damaged cervical tissue	Previous cone/LEEP, repeated dilatation (D&C), obstetric cervical lacerations, congenital uterine anomalies

4. Pathophysiology

The cervix you need to know

```
        Uterine body
            |
     ----- internal os -----      <- cervical stenosis usually here
    |   endocervical canal   |    <- simple columnar, mucus-secreting crypts
    |   (crypts / "glands")  |
    |                        |
    |  original SCJ     new SCJ   <- the TRANSFORMATION ZONE lies between them
     ---- external os -------     <- nulliparous: round; parous: transverse slit
         ectocervix (squamous)
```

- **Length** about 2.5–3 cm; the ectocervix projects into the vaginal vault, surrounded by the anterior, posterior and lateral fornices.
- **Blood supply**: the cervical (descending) branches of the **uterine artery** enter at about **3 and 9 o'clock**: this is why haemostatic sutures after a cone/LEEP are placed at 3 and 9.
- **Ureter**: runs about **1.5–2 cm lateral to the supravaginal cervix**, passing **under the uterine artery** ("water under the bridge"). A cervical fibroid or a cancer in the parametrium distorts or obstructs it.
- **Nerve supply**: the ectocervix has few pain fibres (sensitive to stretch, not to cutting/heat), which is why VIA, biopsy, polypectomy and thermal ablation are done in clinic without anaesthesia.
- **Lymphatics**: to parametrial, obturator, internal/external iliac nodes (important for cancer staging).

Why ectropion and the transformation zone happen

1. At birth, puberty, in pregnancy and on the combined pill, **oestrogen** makes the cervix bulk up and **evert**, exposing endocervical columnar epithelium onto the ectocervix: that exposed columnar tissue is the **ectropion**. It looks red because the single cell layer lets you see the underlying capillaries.
2. The exposed columnar cells now sit in the **acidic vaginal pH (about 4)**. Reserve cells beneath them respond by **squamous metaplasia**: the columnar cells are gradually replaced by squamous cells. The zone where this is happening is the **transformation zone**.

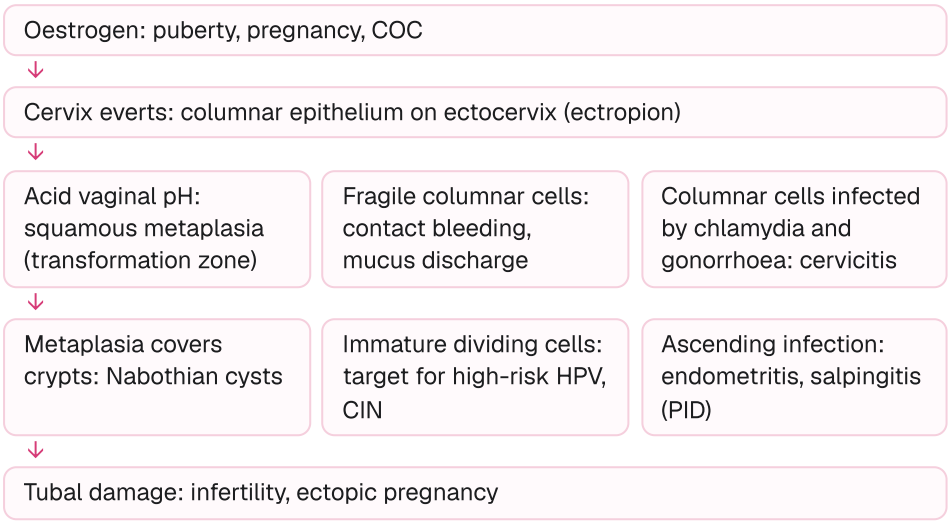
3. Metaplastic squamous epithelium sometimes grows over the mouth of a crypt, trapping mucus → **Nabothian cyst**. So Nabothian cysts mark the outer edge of the TZ.
4. Metaplastic cells are **immature and actively dividing**: the perfect target for **high-risk HPV** to integrate and cause CIN. That is why the TZ is where cancer starts, and why screening (VIA/HPV/cytology) samples the TZ.
5. Exposed columnar epithelium is thin and fragile: it **bleeds on contact** (post-coital bleeding), **produces more mucus** (discharge), and is the preferred target of *Chlamydia* and *N. gonorrhoeae*, which infect **columnar**, not squamous, cells. Ectopy also increases the risk of acquiring HIV.
6. After menopause the SCJ retreats **into the canal** (type 3 TZ), which is why VIA is unreliable after about 50 years and why stenosis is common.

Why cervicitis causes what it causes

- *C. trachomatis* and *N. gonorrhoeae* infect **columnar cells** of the endocervix → neutrophil exudate (**mucopus**), oedema and fragile vessels (**friability, contact bleeding**).
- Organisms can **ascend** through the canal (especially at menstruation, after abortion/delivery or instrumentation) → endometritis → salpingitis → **PID**. Tubal cilia are destroyed and the fimbriae agglutinate → **tubal blockage, hydrosalpinx, ectopic pregnancy and infertility**. Chlamydia in particular does this silently.
- **Trichomonas** causes a vaginitis with a punctate "**strawberry cervix**" (colpitis macularis), not true endocervicitis.
- **Candida and bacterial vaginosis** are **vaginal** infections: they do not cause mucopus from the os. That distinction is the basis of the WHO/Uganda syndromic flowchart: *vaginal* infection vs *cervical* infection.
- **Chronic cervicitis**: long-standing low-grade inflammation, often after childbirth lacerations or repeated infection; the cervix may be hypertrophied with many Nabothian cysts and a mucoid discharge.

Polyps, fibroids and stenosis

- **Endocervical polyps** arise from focal hyperplasia of endocervical glands and stroma; they have a vascular core and so bleed easily (inter-menstrual bleeding, PCB).
- **Cervical fibroids** are smooth-muscle tumours like uterine fibroids, but because the cervix is fixed in the pelvis they cause **pressure effects** (urinary retention/frequency, constipation, dyspareunia), **distort the ureters and bladder**, and can **obstruct labour**. A submucous uterine fibroid can also prolapse through the os and masquerade as a cervical polyp ("**fibroid polyp**").
- **Stenosis** follows scarring (LEEP/cone, cryotherapy, radiotherapy, cervical cancer), or oestrogen deficiency after menopause. Blood or pus cannot drain → **haematometra** (young: cyclic pain, amenorrhoea) or **pyometra** (elderly: offensive discharge, fever), and **pyometra in an elderly woman is cancer (cervix or endometrium) until proven otherwise**.



Feature links

- Exposed columnar epithelium, visible capillaries → red velvety cervix.
- Fragile columnar epithelium → post-coital bleeding, contact bleeding.
- More mucus-secreting columnar cells → mucoid discharge.
- GC/CT infection of columnar cells → neutrophil exudate → mucopus from the os.
- Inflamed fragile vessels → friability, bleeding on swab touch.
- Ascent to the tubes → lower abdominal pain, cervical motion tenderness, fever (PID).
- Tubal cilia destroyed, fimbriae agglutinated → infertility, ectopic pregnancy.
- Metaplasia over crypt openings → Nabothian cysts.
- Vascular core of a polyp → inter-menstrual bleeding, PCB.
- Fibroid fixed in the pelvis → urinary frequency/retention, constipation, dyspareunia, obstructed labour, displaced ureters.
- Stenosed canal → haematometra (cyclic pain, amenorrhoea) or pyometra (offensive discharge, fever).
- Weak cervix → painless second-trimester dilatation → mid-trimester loss.

5. Classification / Staging

A practical classification of benign cervical lesions

1. **Physiological / developmental**: ectropion, Nabothian cysts, squamous metaplasia.
2. **Inflammatory / infective**: acute and chronic cervicitis (gonorrhoea, chlamydia, *Mycoplasma genitalium*, trichomonas, herpes simplex, schistosomiasis, TB (rare)), condylomata (HPV 6/11).
3. **Neoplastic but benign**: polyps, fibroids, (rarely) papillomas and haemangiomas.
4. **Traumatic / iatrogenic**: obstetric lacerations, cervical stenosis after LEEP/cone/cryotherapy/radiotherapy, cervical insufficiency.

> EXAM FAVOURITE
> **TZ types** (colposcopy): **Type 1**: fully ectocervical and fully visible; **Type 2**: has an endocervical component but is fully visible; **Type 3**: extends into the canal and is **not** fully visible. Only type 1 (and some type 2) lesions are suitable for ablation (cryotherapy/thermal ablation); type 3 needs excision (LEEP).

VIA result categories: **negative**; **positive** (dense, well-defined acetowhite lesion at or near the SCJ); **suspicious for cancer** (growth, ulcer).

6. Clinical Features

History

Work through this checklist; every item has an answer that **changes the plan**.

Ask	Why it matters
Main complaint and its timing: discharge? bleeding (post-coital, inter-menstrual, post-menopausal)? pain? lump?	Bleeding after sex or menopause → cancer must be excluded: she needs a speculum look, a screening test and a low threshold for biopsy, not a packet of antibiotics. Discharge alone → infection or ectropion
Discharge: colour, smell, amount, itch, onset, relation to cycle	Curdy + itch = candida; thin grey fishy = BV; frothy green-yellow = trichomonas; mucopurulent/yellow = cervicitis; blood-stained, offensive or watery = **think cancer**
Lower abdominal pain, fever, deep dyspareunia, dysuria	Ascending infection (PID) changes the treatment (14-day regimen, admission if severe)

Ask	Why it matters
Sexual history (risk assessment): new partner in the last 3 months, more than one partner, partner with urethral discharge or genital ulcer, age under about 21–25, condom use	These are the **risk-assessment** questions in the syndromic flowchart: **any "yes" = treat for cervical infection** even if the cervix looks clean
Age	Under 25 years: STIs and ectropion common; over 35 years: polyps, fibroids and cancer more likely
LMP, contraception (COC, IUD, DMPA), possibility of **pregnancy**	COC and pregnancy cause ectropion; IUD strings and infection, IUD matters in PID; pregnancy removes doxycycline and fluconazole from the menu
Cervical screening history: ever had VIA/HPV/Pap? when? result? treated (cryo/thermal/LEEP)?	An **unscreened woman aged 25–49 (or any woman living with HIV)** is screened **today**; prior LEEP/cone → stenosis or insufficiency
HIV status, ART, CD4/last viral load	HIV raises risk of STIs, candida, persistent HPV, CIN and cancer; she gets **screen, triage and treat** and shorter screening intervals
Previous episodes and treatment	A woman who has already failed one syndromic packet needs a **swab for NAAT/culture** and a hunt for another diagnosis, not the same packet again
Obstetric history: parity, mid-trimester losses, cervical tears, recent delivery/abortion/D&C/instrumentation	Chronic cervicitis, lacerations, cervical insufficiency, stenosis, ascending infection
Menstrual history and menopause	Post-menopausal bleeding (PMB) needs **endometrial** as well as cervical assessment
Pressure symptoms: urinary frequency/retention, constipation	Cervical fibroid
Constitutional / "late" symptoms: weight loss, back or sciatic pain, leg swelling, urine or faeces per vagina	Features of **advanced cervical cancer**: do not miss; change pathway
Partner: symptoms, can he come?	Treatment of the couple prevents re-infection

Examination

Follow the order of Hutchison's Clinical Methods (24th edition). If she is bleeding heavily, shocked or septic, resuscitation comes first (section 9).

Introduction, consent and preparation

- Introduce yourself, confirm identity, explain the examination and gain **consent**; a **chaperone** for every speculum and bimanual examination.
- **Hand hygiene** before and after.
- Ask her to **empty her bladder**. **Position and exposure**: supine for the abdomen; dorsal or lithotomy with good light for the pelvic examination; keep her covered except for the area examined.

General inspection and vital signs

- Well or unwell, distress, **pallor** (chronic bleeding), wasting/weight, fever.
- **Vitals**: pulse, BP, temperature, respiratory rate; **shock index** (HR ÷ SBP) if bleeding.

Hands, face and neck

- Palmar and conjunctival pallor; **supraclavicular nodes** (metastatic cervical cancer).

Abdomen and groins

- **Inspection** and **palpation**: tenderness and guarding (PID), a mass arising from the pelvis (fibroid, pyometra, TOA), a palpable bladder (retention from a cervical fibroid), ballotable kidneys (hydronephrosis: think cancer).

- **Inguinal lymph nodes**; leg oedema.

Pelvic examination

1. **Vulva/introitus**: discharge at the introitus, ulcers, warts, excoriation (candida).
2. **Cusco speculum**: the key examination. Warm and lubricate, insert obliquely and rotate. Describe in this order:
   - **(a) Discharge**: amount, colour, and whether it comes **from the os** (cervical, mucopurulent) or coats the **vaginal walls** (vaginal).
   - **(b) The SCJ**: visible, or inside the canal (menopause).
   - **(c) The ectocervix**: ectropion (red, velvety, **symmetrical** area around the os with a **regular, clear border**, not raised, os in the centre); Nabothian cysts (smooth, dome-shaped, yellow/white, translucent blebs); friability (does it bleed when touched with a cotton swab?).
   - **(d) Any mass**: polyp (red/purple, smooth, pedunculated mass protruding from the os; try to see the stalk), fibroid, growth, ulcer. **Growth, ulcer, irregular/necrotic tissue, raised edges, hard texture → suspicious for cancer**.
   - **(e) Vaginal walls**.
3. **Take samples before acetic acid**: endocervical swab (NAAT/culture) and high vaginal swab (wet mount/KOH/pH) where a laboratory exists; HPV sample if HPV-based screening is available.
4. **VIA** if due and she is not pregnant or menstruating (acetic acid 3–5%, read at 1 minute).
5. **Bimanual**: **cervical motion tenderness** (PID), adnexal tenderness or mass, uterine size, a cervix that is **hard, bulky or irregular** (cancer or fibroid), parametrial thickening.
6. **Rectal/rectovaginal** only if cancer is suspected (to assess the parametria).

Completing the examination

Remove the speculum carefully, cover her, thank her, hand hygiene, explain the findings and **summarise**.

Document the key negatives: "no CMT, no adnexal mass, no growth or ulcer, SCJ fully visible, no contact bleeding after treatment". These negatives are what let you call a lesion benign.

> CLINICAL PEARL
> **Ectropion vs early cancer by eye:** ectropion is flat, symmetrical around the os, with a regular border, and turns **pale-pink/transparent (not dense white)** with acetic acid; you can often see grape-like columnar villi. A cancer is raised or ulcerated, irregular, asymmetric, hard, bleeds heavily and may be necrotic. If you cannot be sure, **biopsy**.

> CLINICAL PEARL
> A well-written speculum note makes the diagnosis for you: "Mucopurulent discharge **from the os**, oedematous friable cervix, no CMT" = cervicitis without PID. "Thin grey discharge **coating the vaginal walls**, healthy cervix" = vaginitis. "Smooth red 1 cm pedunculated mass, stalk seen arising from the canal" = polyp.

Red flags

RED FLAG

- **Post-coital, inter-menstrual or post-menopausal bleeding**: screen and biopsy as needed.
- **Visible growth, ulcer, or a hard/irregular cervix**: biopsy the same day (or refer the same week).
- **Offensive blood-stained discharge**, weight loss, back or leg pain, leg swelling, urinary or faecal leakage: advanced cancer.
- **Pyometra in an elderly woman**: cervical or endometrial cancer until proven otherwise.
- **Cervical motion tenderness, fever, peritonism**: PID/tubo-ovarian abscess; exclude **ectopic pregnancy** with a pregnancy test.
- **Symptoms not settling after one full course** of syndromic treatment: do not just repeat the same drugs; examine, screen, test, consider resistance and cancer.

RED FLAG

Call for help at once if: shock index ≥0.9 or signs of shock; heavy active bleeding; a positive pregnancy test with pain, bleeding or collapse (ectopic until proven otherwise); fever with peritonism; or any bleeding in the second half of pregnancy before placenta praevia is excluded.

Thinking like a clinician

- **Start with the "must not miss" and then the "most likely".** In a 45-year-old with PCB, the most likely cause is a polyp or ectropion, but the must-not-miss is cancer. Your plan must address both.
- **Where is the discharge coming from?** From the os = cervicitis (treat GC/CT). From the vaginal walls = vaginitis (BV/TV/candida). This single speculum observation decides the drug combination.
- **"Red cervix" in a 19-year-old on COC** with a regular-bordered velvety area is ectropion; test/treat for chlamydia (ectopy is its favourite target), confirm she is not overdue for screening (in Uganda screening usually starts at 25–30 unless HIV-positive), and **reassure**. Do not cauterise to "cure" a normal variant.
- **Pregnant woman with PCB**: ectropion is very common in pregnancy; but still speculum (after placenta praevia has been excluded by ultrasound in the late second/third trimester), and remember cervical cancer can occur in pregnancy.
- **HIV-positive woman with recurrent discharge**: think of STIs, candida, but also CIN/cancer: she needs screening (HPV/VIA) at the visit.
- **A "polyp" that bleeds a lot, is broad-based or is fleshy/firm** may be a prolapsed submucous fibroid or a polypoid cancer: do not avulse blindly in OPD; feel for the stalk, and arrange ultrasound/theatre.
- **Heavy bleeding from the cervix** is far more often a **cancer, an incomplete miscarriage or a prolapsed fibroid** than a polyp or ectropion.
- **Treat the couple, not just the woman.** If her partner is not treated, she is re-infected in weeks ("ping-pong" infection). Give partner notification slips.
- **Screening tests are for asymptomatic women**: a symptomatic woman with persistent PCB despite a normal screen needs diagnostic assessment (colposcopy ± biopsy).
- **Common student mistakes:** calling every red cervix "cervicitis"; doing a speculum without looking for the SCJ; forgetting a pregnancy test before doxycycline; forgetting partner treatment and HIV testing; avulsing a polyp and not sending it to histology; reassuring a woman with PCB without a screening test.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Ectropion	Young, pregnant or on COC; red velvety symmetrical area around os, regular border; mucoid discharge, PCB	Speculum + VIA/HPV screen; test for chlamydia if possible
Acute cervicitis (GC/CT)	Mucopurulent discharge from os, oedematous friable cervix, contact bleeding; partner symptoms; ± dysuria	Endocervical NAAT (GeneXpert CT/NG) or Gram stain/culture where available; syndromic otherwise
PID	Cervicitis **plus** lower abdominal pain, cervical motion/adnexal tenderness, fever	Pregnancy test, FBC/CRP, pelvic ultrasound (TOA)
Vaginitis (BV/TV/candida)	Discharge on vaginal walls; itch (candida), fishy odour (BV), frothy + strawberry cervix (TV)	Wet mount, KOH/whiff test, vaginal pH
Cervical polyp	Smooth red pedunculated mass at os; IMB/PCB	Avulsion + histology
Prolapsed submucous fibroid	Firm, pale mass through os, stalk from the uterine cavity, heavy bleeding, cramping	Pelvic ultrasound; examination under anaesthesia
Nabothian cysts	Smooth translucent yellow-white blebs, asymptomatic	Clinical (ultrasound if deep)
Cervical fibroid	Bulky, firm, smooth cervix; pressure symptoms; uterus pushed up ("lantern on the dome")	Pelvic ultrasound
CIN / early cervical cancer	Often asymptomatic; PCB; acetowhite lesion at SCJ; friable, irregular growth	VIA/HPV, colposcopy, **punch biopsy**
Advanced cervical cancer	Offensive bloody discharge, hard fixed cervix, parametrial involvement, fistula	Biopsy; staging (FIGO)
Condylomata acuminata	Warty growths on cervix/vulva (HPV 6/11); HIV	Clinical; biopsy if atypical
Genital schistosomiasis / TB	Endemic areas; "sandy patches", grainy lesions; infertility	Biopsy; urine/stool ova, TB workup
Endometrial pathology	Post-menopausal bleeding with normal cervix	TVUS endometrial thickness, endometrial biopsy
Ectopic pregnancy / miscarriage	Positive pregnancy test with pain or bleeding	Urine hCG, ultrasound
Placenta praevia	Painless bleeding after about 20 weeks	Ultrasound **before** any speculum

8. Clinical Workup

This is the order in which you would work up a woman who comes to the gynae clinic (or the OPD at an HC IV) with **discharge, post-coital or inter-menstrual bleeding, or "something on the cervix"**. Most of these women are well and walking; the workup is about **not missing cancer, PID, ectopic pregnancy or placenta praevia**, and then naming the benign lesion precisely. The first 5 minutes (vitals, heavy bleeding, pregnancy, sepsis) are in section 9.

Bedside

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Test	Minutes	What it tells you	Available at
Urine pregnancy test (hCG)	3–5	Must be done before doxycycline, before any procedure and in every woman with pain/bleeding (ectopic)	All levels
VIA	2	Negative / positive (dense acetowhite at the SCJ) / suspicious for cancer (growth, ulcer). Same-visit treatment decisions	HC III (where trained) and above
HemoCue / Hb	2	Anaemia from chronic bleeding (polyp, fibroid, cancer); Hb <7 g/dL = severe	HC III and above
Vaginal pH strip (where available)	1	**>4.5** suggests BV or trichomonas; **≤4.5** with curdy discharge suggests candida	Variable
Whiff test (10% KOH on the discharge)	1	Fishy amine smell = BV (also trichomonas)	Where KOH is available
Wet mount / KOH microscopy (if a microscope is on the ward)	5–10	Clue cells (BV), motile trichomonads, hyphae/spores (candida), many pus cells	HC IV labs with a microscope
Rapid HIV and syphilis tests	15	Any STI is a marker for others; HIV changes screening and follow-up	All levels
Random blood sugar	1	Recurrent candida: exclude diabetes	HC III and above
Point-of-care ultrasound (if skilled)	5	Free fluid/adnexal mass in suspected ectopic/TOA; placenta site before speculum in late pregnancy	HC IV and above with an operator

Reading vaginal microscopy

Finding	What you are looking for	Why
Bacterial vaginosis (Amsel criteria, 3 of 4)	Thin, grey, homogeneous discharge; **pH >4.5**; **positive whiff test** (fishy smell with 10% KOH); **clue cells** (epithelial cells coated with bacteria) on wet mount	Metronidazole; BV is a vaginal, not cervical, infection
Trichomoniasis	Motile flagellated trichomonads on a fresh saline wet mount (read within minutes); pH >4.5; "strawberry cervix"	Metronidazole for her **and** her partner
Candida	Hyphae and spores on KOH mount; pH usually ≤4.5	Clotrimazole (pregnancy) or fluconazole; recurrent → check glucose and HIV
Gonorrhoea on Gram stain	Gram-negative **intracellular diplococci** in pus cells (less sensitive in women than in men)	A negative Gram stain does **not** exclude cervical gonorrhoea: keep treating syndromically
Many pus cells, no organism	Cervicitis (chlamydia is not seen on microscopy)	Treat for GC + CT if mucopus/risk assessment positive

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
High vaginal swab: wet mount, KOH, Gram (Nugent)	Vaginal discharge, recurrent or unresponsive symptoms	Clue cells, trichomonads, yeast	Chooses metronidazole vs antifungal vs both	HC IV labs with a microscope
FBC	Bleeding, suspected PID	Anaemia; leucocytosis	Iron or transfusion; supports severe PID	HC IV and above
Syphilis (RPR/TPHA), hepatitis B	Any STI syndrome	Positive serology	Treat and trace partners	HC III/IV (rapid tests) and above

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Pelvic ultrasound (TVUS/TAS)	Cervical/prolapsed fibroid, pyometra/haematometra, suspected TOA, PMB	Fibroid size and site; fluid in the cavity; adnexal mass; **endometrial thickness** in PMB	Plans vaginal myomectomy vs hysterectomy; PMB with a thick endometrium → endometrial sampling	HC IV (basic) and above
Obstetric ultrasound	Bleeding after about 20 weeks	Placental site	Excludes placenta praevia before any speculum	HC IV and above

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Endocervical swab: NAAT for CT/NG (GeneXpert)	Mucopus from the os, positive risk assessment, failed syndromic treatment; any woman where available	Positive CT and/or NG	Confirms cervicitis; targeted treatment and partner tracing	GeneXpert CT/NG cartridges mainly at regional/national referral hospitals and research sites; **not routine at HC III/IV**, hence syndromic care
Gram stain / culture of endocervical swab	Suspected gonorrhoea, especially after treatment failure	Gram-negative intracellular diplococci; culture gives **sensitivities**	With Ugandan AMR to cefixime, azithromycin and doxycycline, a sensitivity result directs second-line therapy	Regional/national referral labs
HPV DNA test (clinician or self-sample)	Every woman due for screening (general population from 30, WLWH from 25)	HPV negative / positive (± 16/18)	Negative = very low risk, re-screen at interval; positive → **VIA triage** and treatment	GeneXpert hubs and HIV clinics; sample can be taken at HC III/IV and transported
Punch biopsy	Any growth, ulcer, suspicious or atypical lesion; persistent PCB with an abnormal-looking cervix	CIN or invasion vs benign inflammation	Only histology diagnoses cancer	Biopsy at HC IV/general hospital; histology read at regional/national referral labs
Colposcopy	Persistent PCB with negative screening; suspected glandular lesion; SCJ not seen	Acetowhite, mosaicism, punctation, atypical vessels	Guides biopsy; decides ablation vs LEEP	Referral hospitals (Mengo, Mulago, UCI, regional referrals)

Test	When	Expected / abnormal finding	How it changes management	Available at
Polyp histology	Every polyp removed	Benign endocervical polyp (expected) vs dysplasia/malignancy (rare)	Malignant or dysplastic → gynae-oncology referral	Regional/national referral labs
Endometrial biopsy (Pipelle/D&C)	PMB with thick endometrium or persistent bleeding	Hyperplasia/cancer	Changes the whole diagnosis	General/referral hospital

> **UGANDA CONTEXT**
> At **HC III/IV** your "lab" is the speculum, VIA, a pregnancy test, rapid HIV/syphilis tests and perhaps a microscope: that is exactly why Uganda uses **syndromic management**. **NAAT, culture and sensitivities, colposcopy and histology** live at general, regional and national referral hospitals. Refer or send a sample when the woman has **failed one full course**, has **suspicious findings**, or has **persistent PCB**.

Putting it together

Model one-line summary: "A 34-year-old para 3, HIV-negative, last screened never, with 2 months of post-coital spotting; speculum shows a 1 cm smooth pedunculated mass at the os with the stalk seen in the canal, the rest of the cervix VIA-negative, no CMT; pregnancy test negative, Hb 11.8 g/dL."

Problem list

1. Post-coital bleeding (cancer must be excluded).
2. Mass at the os: likely endocervical polyp.
3. Never screened.
4. Sexual health: risk assessment, HIV and syphilis status.

Working diagnosis: endocervical polyp, with VIA negative today; differentials: prolapsed submucous fibroid, polypoid cancer, ectropion.

Plan: avulse the polyp in clinic, send it for histology, HIV/syphilis tests, condoms/counselling, review with histology in 2–4 weeks; if PCB persists after polypectomy → colposcopy.

Model case presentation 1 (clinic)

> "Mrs N, a 38-year-old P4, presents with a 3-month history of post-coital bleeding. She has never been screened and is HIV-negative. On speculum there is a 1 cm smooth red pedunculated mass arising from the os; the rest of the cervix appears healthy. My impression is an endocervical polyp; differentials are a prolapsed submucous fibroid, ectropion and cervical cancer. I would do VIA/HPV testing, avulse the polyp and send it for histology, and review with results."

Model case presentation 2 (ward round)

> "A 26-year-old para 1+1, HIV-negative, with a 2-week history of yellow vaginal discharge and post-coital spotting, a new partner 2 months ago and a partner with dysuria. Speculum shows mucopurulent discharge from a friable oedematous cervix; no cervical motion tenderness. Impression: acute cervicitis, likely

> gonococcal/chlamydial, without PID. Plan: pregnancy test, endocervical swab for NAAT if available, ceftriaxone 1 g IM stat (or cefixime 400 mg), doxycycline 100 mg 12-hourly for 7 days and metronidazole 2 g stat; partner notification and treatment, condoms, HIV/syphilis testing, VIA once treated, review in 7 days."

Worked example 1: reading a speculum + VIA result

> A 29-year-old woman living with HIV (viral load suppressed) attends for screening. HPV test: **positive**. Speculum: SCJ fully visible; after acetic acid, a **dense, well-defined acetowhite area at the SCJ from 11 to 2 o'clock**, covering about a quarter of the ectocervix, not entering the canal; no growth or ulcer. Pregnancy test negative.

- **Interpretation**: HPV-positive, **VIA-positive** (triage positive), lesion **eligible for ablation** (whole lesion and SCJ seen, <75% of the ectocervix, no canal extension, no suspicion of cancer, not pregnant).
- **Next step**: **thermal ablation the same day** (≥100 °C, 20–30 s per application, overlapping) or cryotherapy (3-5-3). Advise watery discharge for 2–4 weeks and no intercourse/tampons for 4 weeks; **re-test at 12 months**.
- **Contrast**: the same woman with a raised, irregular, ulcerated area that bleeds on touch = **"suspicious for cancer" → punch biopsy from the edge and refer**; **never ablate**. A flat, symmetrical, red area around the os that turns only pale pink/translucent = **ectropion**, VIA negative.

Worked example 2: reading discharge findings

> A 22-year-old with yellow discharge; new partner 1 month ago; partner has dysuria. Speculum: **mucopus from the os**, friable cervix; no CMT. pH 5.0, whiff test positive, wet mount shows clue cells. Pregnancy test negative.

- **Interpretation**: **cervicitis** (mucopus + positive risk assessment) **plus BV** (Amsel: pH >4.5, positive whiff, clue cells). No CMT, so **no PID**.
- **Next step**: **ceftriaxone 1 g IM stat (or cefixime 400 mg PO stat) + doxycycline 100 mg PO 12-hourly for 7 days + metronidazole** (2 g stat, or 400–500 mg 12-hourly for 7 days, which cures BV better); partner slip and partner treatment, condoms, HIV/syphilis tests, **review in 7 days** and do VIA/HPV once the inflammation settles.
- **If she were pregnant**: azithromycin 1 g stat instead of doxycycline; metronidazole 400–500 mg 12-hourly for 7 days.

Worked example 3: post-menopausal "polyp"

> A 58-year-old with PMB has a small cervical polyp. After avulsion, **TVUS shows an endometrial thickness of 11 mm**.

- **Interpretation**: the polyp may not be the (only) source; a thick endometrium in PMB needs **endometrial sampling**. Most units use **>4 mm** as the cut-off for sampling in PMB (some use 5 mm; follow your unit).
- **Next step**: Pipelle or D&C (ideally hysteroscopy) for histology before any reassurance.

9. Management

Immediate / emergency

Most women are stable, but check before you reach for the speculum (first 5 minutes):

1. **Vitals (A, B, C)**: pulse, BP, temperature, respiratory rate. Work out the **shock index** (HR ÷ SBP) if she is bleeding; a shock index of 0.9 or more, pallor, cold peripheries or dizziness means she is **not** an OPD patient.

2. **Is she bleeding heavily now?** (soaking pads, clots, faintness) → lie her down, **IV access (two large-bore cannulae)**, **Hb and group and cross-match**, start crystalloid and **call a senior**. Heavy bleeding from the cervix is far more often a **cancer, an incomplete miscarriage or a prolapsed fibroid** than a polyp or ectropion.
3. **Could she be pregnant?** Ask the LMP and do a **urine pregnancy test** on everyone of reproductive age. Positive + pain/bleeding → think **ectopic pregnancy or miscarriage** first. Positive and beyond about 20 weeks with bleeding → **no speculum or digital examination until an ultrasound has excluded placenta praevia**.
4. **Is she septic or peritonitic?** Fever ≥38.0 °C, tachycardia, guarding/rebound, vomiting → treat as **severe PID / tubo-ovarian abscess** (or a surgical abdomen): admit, IV access, bloods, IV antibiotics, and call the team.
5. **Obvious advanced cancer** (fungating, foul-smelling growth, fistula, uraemic) → this is now the malignant pathway: biopsy, stage, refer and palliate.

Definitive

Principles (apply to every benign cervical lesion)

1. **Exclude pregnancy and malignancy**: speculum, screening test (VIA/HPV) if due, biopsy of anything suspicious.
2. **Identify and treat infection** (syndromically or by test results), and **treat partners**.
3. **Treat the lesion only if it causes symptoms** or histology is needed (polyp).
4. **Counsel**: explain that the finding is benign, give condoms, offer HIV testing, and arrange screening follow-up.

A. Syndromic management of vaginal discharge (HC II–IV)

This is how a Ugandan health worker manages a woman who says "I have a discharge":

1. **History + risk assessment.** Risk assessment is **positive** if **any** of: partner has urethral discharge/genital ulcer; she is young (under about 21–25) or has had a new partner in the last 3 months or more than one partner.
2. **Examine** (abdomen, vulva, speculum ± bimanual).
3. **Lower abdominal pain or cervical motion tenderness?** → manage as **PID** (B, below).
4. **Mucopurulent cervical discharge, friable cervix, or positive risk assessment?** → treat **cervical infection (gonorrhoea + chlamydia) and** vaginal infection (BV + trichomonas).
5. **Vaginal discharge only, risk assessment negative** → treat **vaginal infection** (BV + trichomonas); **add candida treatment** if curdy discharge / itch / vulval erythema.
6. **Always**: 4 Cs (compliance, counselling, condoms, contact tracing), HIV and syphilis testing, and **review in 7 days**.

Cervical infection (cervicitis), gonorrhoea + chlamydia: **cefixime 400 mg PO stat** (UCG option) **or ceftriaxone 1 g IM stat** (WHO 2024 preferred single-dose regimen because of rising cefixime/azithromycin resistance) **plus doxycycline 100 mg PO 12-hourly for 7 days**. **In pregnancy**: **azithromycin 1 g PO stat** replaces doxycycline (or erythromycin 500 mg PO 6-hourly for 7 days).

Vaginal infection, BV + trichomonas (± candida): **metronidazole 2 g PO stat** (covers trichomonas and BV) or **400–500 mg PO 12-hourly for 7 days** (better cure for BV; preferred in pregnancy and for recurrence). Candida: **clotrimazole 500 mg pessary** at night once (or 100 mg nightly for 6 nights / 200 mg for 3 nights; safe in pregnancy) or **fluconazole 150 mg PO stat** (avoid in pregnancy; for recurrent candida consider longer courses and check glucose/HIV). Doses in the drug table.

> **UGANDA CONTEXT**
> The Uganda Clinical Guidelines use **cefixime 400 mg stat or ceftriaxone** + **doxycycline 100 mg 12-hourly for 7 days** (azithromycin/erythromycin in pregnancy) + **metronidazole 2 g stat** as the core package for a woman with abnormal discharge and cervical infection. WHO's 2024 update prefers **ceftriaxone 1 g IM** as single-dose gonorrhoea treatment because of antimicrobial resistance. Ugandan resistance data (doxycycline, azithromycin and cefixime) mean that **a woman who fails one course needs a swab for culture/NAAT**, not another identical packet.

Partner management: give a notification slip; treat the partner(s) with the **urethral discharge regimen** (ceftriaxone/cefixime + doxycycline) whether or not he has symptoms; **abstain or use condoms until 7 days after both have completed treatment**.

B. Pelvic inflammatory disease (when cervicitis has ascended)

Features: lower abdominal pain + cervical motion/adnexal tenderness ± fever, ± mucopurulent discharge. **Always do a pregnancy test** (ectopic!).

- **Outpatient (mild–moderate; can tolerate oral drugs)**, international regimen (CDC/BASHH, used in Uganda): **ceftriaxone 1 g IM stat** (CDC: 500 mg if <150 kg) + **doxycycline 100 mg PO 12-hourly for 14 days** + **metronidazole 400–500 mg PO 12-hourly for 14 days**.
- **Admit (HC IV / hospital) if**: severe illness, high fever, vomiting, peritonism, tubo-ovarian abscess, pregnancy, failed oral therapy after 72 h, or diagnosis unclear (appendicitis, ectopic).
- **Inpatient**: **ceftriaxone 1 g IV daily + metronidazole 500 mg IV 8-hourly + doxycycline 100 mg PO 12-hourly**, switching to oral once improved for 24–48 h to complete 14 days. Remove an IUD only if no improvement in 48–72 h. Drain a TOA that does not respond.

> **NOTE**
> Check the exact PID regimen in the current Uganda Clinical Guidelines at your facility: the principle (a cephalosporin for gonorrhoea + doxycycline for chlamydia + metronidazole for anaerobes, **14 days**) is universal.

C. Ectropion

1. Confirm the diagnosis on speculum; **screen** (VIA/HPV) if due; test/treat for chlamydia if symptomatic or at risk.
2. **Asymptomatic** → reassure; no treatment. Explain it is a normal variant that often regresses.
3. **Symptomatic** (troublesome mucoid discharge or PCB) **after** normal screening and treated infection:
   - If on COC and symptoms bother her, consider switching to a progestogen-only method (reduces oestrogen stimulus).
   - **Ablation** (cryotherapy, thermal ablation (cold coagulation) or diathermy) in clinic; warn of watery discharge for 2–4 weeks, and avoid intercourse/tampons for about 4 weeks.
4. **In pregnancy**: reassure; it regresses postpartum. Never ablate in pregnancy.

D. Nabothian cysts

- **No treatment** if asymptomatic (almost always).
- Large symptomatic cysts (rare): puncture/drain or ablate. If the cysts are deep, multiple and the cervix is hard or enlarged, get an ultrasound and consider **adenoma malignum** (rare minimal-deviation adenocarcinoma): biopsy.

E. Cervical polyp: step by step (OPD procedure)

1. Counsel and consent; exclude pregnancy; screen the cervix (VIA/HPV).

2. Speculum; clean with saline; identify the pedicle.
3. Grasp the polyp at its base with **sponge-holding (polyp) forceps** and **twist (avulse)** repeatedly in one direction until it detaches.
4. Control bleeding from the base with pressure, **silver nitrate** or Monsel's solution; diathermy if needed.
5. Put the specimen in **formalin** and send for **histology**: always.
6. **Refer to theatre** (EUA/hysteroscopic removal) if the base is broad (sessile), the polyp is large or very vascular, the stalk arises high in the canal or from the uterine cavity (prolapsed fibroid), or if it looks suspicious.
7. **Post-menopausal woman** (or abnormal bleeding persists): also assess the **endometrium** (TVUS endometrial thickness ± endometrial sampling), because coexisting endometrial pathology is common.
8. Review with histology in 2–4 weeks.

F. Cervical fibroid

1. **Ultrasound** to map it; FBC; correct anaemia (ferrous sulphate 200 mg PO 8-hourly).
2. **Asymptomatic, small** → observe.
3. **Pedunculated fibroid prolapsing through the os ("fibroid polyp")** → vaginal myomectomy: clamp/ligate the pedicle and twist off, in theatre (spinal or GA), with histology.
4. **Symptomatic intramural/supravaginal cervical fibroid** → myomectomy (if fertility desired) or total hysterectomy at a **referral hospital** by an experienced surgeon: the ureters are displaced laterally/upwards and are at high risk; consider pre-operative GnRH analogue where available to shrink it.
5. **In pregnancy**: may obstruct labour → plan **caesarean section**; do not remove at CS (bleeding).

G. Cervical stenosis

1. Ultrasound for haematometra/pyometra.
2. **Dilate** the canal with Hegar dilators under anaesthesia (ultrasound-guided if difficult); drain pus/blood; send endometrial/cervical tissue for histology.
3. **Pyometra**: drainage + antibiotics (e.g. ceftriaxone + metronidazole) + **exclude malignancy** (endometrial and cervical biopsy).
4. Recurrent stenosis may need repeated dilatation or, in post-menopausal women with recurrent pyometra, hysterectomy.

H. Cervical insufficiency (brief)

- **History**: painless second-trimester loss or preterm birth, often with bulging membranes and little pain; risk factors in section 3.
- **History-indicated cerclage** at **12–14 weeks** if **three or more** previous mid-trimester losses/preterm births (RCOG).
- **Ultrasound-indicated cerclage** if a previous mid-trimester loss/preterm birth **and** transvaginal cervical length **≤25 mm before 24 weeks**.
- **Techniques**: McDonald (purse-string at the cervico-vaginal junction; commonest in Uganda) or Shirodkar (higher, after bladder dissection). **Remove at 36–37 weeks**, or immediately in labour, PPROM with infection or bleeding.

> **UGANDA CONTEXT**
> Cerclage in Uganda is done at **general and regional/national referral hospitals** (Mengo, Mulago, regional referral hospitals), usually McDonald under spinal anaesthesia. Transvaginal cervical-length screening is not routine; decisions are usually **history-based**. Vaginal progesterone is an alternative for a short cervix where available.

I. Approach to post-coital bleeding (algorithm)

1. **History**: age, screening history, HIV, contraception, pregnancy, other bleeding, discharge, partner.
2. **Pregnancy test**; if pregnant in the second/third trimester, **ultrasound for placenta praevia before any speculum**.
3. **Speculum**:
   - Obvious growth/ulcer → **biopsy now/refer urgently** (gynae-oncology).
   - Polyp → avulse + histology.
   - Cervicitis → swabs + syndromic treatment; **re-examine after treatment**.
   - Ectropion or normal cervix → **screening test** (VIA/HPV) and STI tests.
4. **VIA/HPV positive** → screen-and-treat pathway (ablation or LEEP; see Malignant lesions of the cervix).
5. **Persistent PCB** despite normal screen and treated infection → **colposcopy ± biopsy** at a referral centre (screening tests are for asymptomatic women; a symptomatic woman needs diagnostic assessment).
6. **Post-menopausal** → also TVUS endometrium ± endometrial biopsy.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Cefixime (gonorrhoea)	400 mg	PO	Single dose	Stat	UCG option; Ugandan resistance reported
Ceftriaxone (gonorrhoea; PID)	1 g (CDC PID: 500 mg if <150 kg)	IM (outpatient); IV daily for inpatient PID	Single dose (cervicitis, outpatient PID); daily (inpatient PID)	Stat; inpatient until improved 24–48 h	WHO 2024 preferred for gonorrhoea; IM in 3.5 mL of 1% lignocaine reduces pain; do not mix with calcium-containing fluids IV
Doxycycline (chlamydia; PID)	100 mg	PO	12-hourly	**7 days** (cervicitis); **14 days** (PID)	**Not in pregnancy or breastfeeding** (tooth discolouration, bone effects): pregnancy test first
Azithromycin (chlamydia in pregnancy)	1 g	PO	Single dose	Stat	Replaces doxycycline in pregnancy
Erythromycin (chlamydia in pregnancy, alternative)	500 mg	PO	6-hourly	7 days	GI upset
Metronidazole (BV, trichomonas)	2 g	PO	Single dose	Stat	Disulfiram-like reaction with alcohol: avoid for 24–48 h
Metronidazole (BV, recurrence, pregnancy; PID)	400–500 mg	PO	12-hourly	7 days (BV); **14 days** (PID)	Better BV cure than 2 g stat; avoid alcohol
Metronidazole (inpatient PID)	500 mg	IV	8-hourly	Until oral tolerated	Switch to oral to complete 14 days
Clotrimazole (candida)	500 mg pessary (or 100 mg × 6 nights / 200 mg × 3 nights)	Vaginal	At night	Single dose (or as stated)	Safe in pregnancy
Fluconazole (candida)	150 mg	PO	Single dose	Stat	**Avoid in pregnancy**; recurrent: longer courses, check glucose/HIV

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ferrous sulphate	200 mg	PO	8-hourly	Until Hb corrected	Anaemia from bleeding (fibroid, polyp)
Silver nitrate / Monsel's solution	Topical to polyp base	Topical	Once	At polypectomy	Haemostasis

DRUG DOSING

Doxycycline is contraindicated in pregnancy (tooth discolouration, bone effects): always ask about the LMP or do a pregnancy test before prescribing it. Metronidazole with alcohol causes a disulfiram-like reaction. Ceftriaxone should be given **IM** in the outpatient setting (1 g in 3.5 mL of 1% lignocaine reduces pain) and should not be mixed with calcium-containing fluids if given IV.

Referral criteria

Referral triggers compiled from the source notes and UCG levels of care Verify *not an MoH referral list*.

Level	What to do	Refer up when
HC III	History, speculum, VIA (where trained), syndromic treatment, partner slips, HIV/syphilis tests	Suspicious lesion, polyp, failed syndromic course, PID needing admission, persistent PCB → HC IV
HC IV	Polypectomy, cryotherapy/thermal ablation, HPV sample collection, mild PID, admission for severe PID, basic ultrasound	Not eligible for ablation (needs LEEP), suspicious lesion for biopsy/histology, colposcopy, sessile or high polyp, cervical fibroid, stenosis/pyometra, TOA not responding, cerclage → General/RRH
General / regional referral hospital	Colposcopy, LEEP, punch biopsy + histology, cultures/NAAT, EUA, vaginal myomectomy, cerclage, hysterectomy for fibroids, drainage of TOA/pyometra	Histology shows dysplasia/malignancy, suspected invasive cancer → NRH/UCI
National referral / UCI	Gynae-oncology	n/a

Monitoring and follow-up

- **Syndromic treatment**: review in **7 days**; abstain or use condoms until 7 days after both partners complete treatment.
- **Failure of one full course**: check adherence and partner treatment, re-examine, **swab for NAAT/culture and sensitivities**, consider resistance and other diagnoses (cancer, TB, schistosomiasis), HIV.
- **Outpatient PID**: reassess at 72 h; admit if no improvement.
- **After polypectomy**: review with histology in 2–4 weeks; colposcopy if PCB persists.
- **After ablation**: watery discharge 2–4 weeks; no intercourse or tampons for 4 weeks; **re-test at 12 months**.
- **Cerclage**: remove at 36–37 weeks, or at once in labour, PPROM with infection, or bleeding.
- **Screening**: do VIA/HPV once inflammation settles; arrange routine screening at the correct interval (section 14).

10. Complications

Disease-related

Condition	Complications
Cervicitis (untreated)	PID, tubo-ovarian abscess, **tubal infertility**, **ectopic pregnancy**, chronic pelvic pain; in pregnancy: PROM, preterm labour, chorioamnionitis, **neonatal ophthalmia** and chlamydial pneumonia; increased HIV transmission/acquisition
Ectropion	PCB, discharge; increased susceptibility to chlamydia/HIV
Polyp	Bleeding, anaemia; rare malignancy; recurrence (about 6%)
Cervical fibroid	Urinary retention, hydronephrosis, obstructed labour, haemorrhage
Stenosis	Haematometra, pyometra, dysmenorrhoea, infertility, inability to sample the TZ/endometrium
Cervical insufficiency	Mid-trimester loss, preterm birth

Treatment-related

Treatment	Complications
Ablation of ectropion (cryotherapy, thermal, diathermy)	Watery discharge, stenosis (rare)
Polypectomy	Bleeding from the base (pressure, silver nitrate, diathermy)
Surgery for a cervical fibroid	**Ureteric injury**, haemorrhage
Cerclage	Rupture of membranes, infection, bleeding, cervical tear if left in during labour
LEEP/cone, cryotherapy, radiotherapy	Cervical stenosis, cervical insufficiency
Drugs	Doxycycline in pregnancy (fetal teeth/bone); metronidazole with alcohol (disulfiram-like reaction); fluconazole in pregnancy

11. Prognosis

- **Ectropion** is a normal variant that often regresses (and regresses postpartum in pregnancy).
- **Nabothian cysts** are harmless.
- **Polyps**: cured by avulsion; recurrence about 6%; malignant change well under 1–2%.
- **Cervicitis**: curable with correct treatment of the woman **and her partner**; untreated, about 10–15% of GC/CT infections ascend to PID, with long-term risk of tubal infertility, ectopic pregnancy and chronic pelvic pain. Rising antimicrobial resistance in Uganda makes follow-up essential.
- **Cervical fibroids**: benign; the risk lies in pressure effects, obstructed labour and surgical (ureteric) injury.

12. Prevention

- **STIs**: condoms, partner notification and treatment, the **4 Cs** (compliance, counselling, condoms, contact tracing), HIV and syphilis testing at every STI visit.
- **Preventing PID and its sequelae**: early syndromic treatment of cervicitis, treating partners, safe abortion and delivery practices (ascending infection after abortion/delivery/instrumentation).
- **Cervical screening** (VIA/HPV) at every opportunity, particularly when a woman attends with cervical symptoms; screen-and-treat to prevent cancer (see Malignant lesions of the cervix).
- **Iatrogenic stenosis and insufficiency**: avoid unnecessary cautery of a physiological ectropion and unnecessary repeated cervical dilatation.

13. Special Populations

- **Pregnancy**: no doxycycline (azithromycin 1 g stat or erythromycin), no fluconazole (clotrimazole), metronidazole 400–500 mg 12-hourly for 7 days; **never ablate** an ectropion; bleeding after about 20 weeks needs ultrasound for placenta praevia before any speculum; PID in pregnancy is an admission; a cervical fibroid may obstruct labour (plan CS, do not remove at CS); cervical cancer can occur in pregnancy.
- **Women living with HIV**: more STIs, candida, persistent HPV, CIN and cancer; screening starts at **25** years with shorter intervals (HPV every 3–5 years); screen at the visit; recurrent candida → check glucose.
- **Post-menopausal women**: the SCJ retreats into the canal (type 3 TZ), so VIA is unreliable after about 50 years; stenosis and pyometra (cancer until proven otherwise); PMB needs endometrial assessment (sample if endometrium >4 mm).
- **Adolescents and young women (under 25)**: ectropion and STIs common; risk assessment positive by age alone.

14. Key Points

High-yield points

- A benign cervical diagnosis is a **diagnosis of exclusion**: look, screen, biopsy anything suspicious.
- The **transformation zone** is where metaplasia, Nabothian cysts and almost all cervical cancers arise.
- **Ectropion** is physiological: treat only if symptomatic **after** normal screening and treated infection; never in pregnancy.
- Mucopus **from the os** = gonorrhoea/chlamydia; discharge **on the vaginal walls** = BV/TV/candida.
- **Any positive risk-assessment answer** = treat for cervical infection even if the cervix looks normal.
- **Pregnancy test before doxycycline**, fluconazole or any procedure; ultrasound before a speculum for bleeding after about 20 weeks.
- **Every polyp goes for histology**; in PMB assess the endometrium too.
- **Treat the partner**: otherwise "ping-pong" re-infection.
- **Failed one course** = swab for NAAT/culture and sensitivities, not the same packet.
- A cervical fibroid displaces the **ureters** (1.5–2 cm lateral, under the uterine artery).

Numbers & doses

Item	Value
VIA	**3–5% acetic acid**, read after **1 minute**; positive = dense, well-defined acetowhite lesion at/near the SCJ
Screening start (WHO 2021)	General population **30** years (HPV DNA every **5–10 years**; VIA every 3 years); women living with HIV **25** years (HPV every **3–5 years**). Uganda intervals are in transition: check the current MoH algorithm
Ablation eligibility	Whole lesion and SCJ visible, lesion **<75%** of the ectocervix, no canal/vaginal extension, no suspicion of cancer, not pregnant
Thermal ablation / cryotherapy	**≥100 °C for 20–30 s** per application / **freeze 3 min, thaw 5, freeze 3**
After ablation	Watery discharge **2–4 weeks**; no intercourse or tampons for **4 weeks**; re-test at **12 months**
Cervicitis (UCG)	**Cefixime 400 mg PO stat** or **ceftriaxone 1 g IM stat** + **doxycycline 100 mg PO 12-hourly × 7 days**
Chlamydia cover in pregnancy	**Azithromycin 1 g PO stat** (or erythromycin 500 mg PO 6-hourly × 7 days)
BV / trichomonas	**Metronidazole 2 g PO stat** or **400–500 mg PO 12-hourly × 7 days**

Item	Value
Candida	**Clotrimazole 500 mg pessary stat** (safe in pregnancy) or **fluconazole 150 mg PO stat** (avoid in pregnancy)
Outpatient PID	Ceftriaxone **1 g IM stat** + doxycycline **100 mg 12-hourly × 14 days** + metronidazole **400–500 mg 12-hourly × 14 days**
PID oral failure	Admit if no improvement after **72 h**
Abstinence after STI treatment	Until **7 days** after both partners complete treatment
Vaginal pH	Normal about **4**; **>4.5** = BV or trichomonas
BV (Amsel)	**3 of 4**: thin homogeneous discharge, pH >4.5, positive whiff test, clue cells
Untreated GC/CT → PID	About **10–15%**
Cervical polyp	Prevalence about **2–5%**; malignant change **well under 1–2%**; recurrence about **6%**
Cervical fibroids	About **1–2%** of all fibroids
Ureter	About **1.5–2 cm lateral** to the supravaginal cervix, **under** the uterine artery
Cervical branches of the uterine artery	**3 and 9 o'clock** (haemostatic sutures)
Cerclage	History-indicated at **12–14 weeks** if **≥3** mid-trimester losses/PTB; ultrasound-indicated if prior loss/PTB **and** cervix **≤25 mm before 24 weeks**; remove at **36–37 weeks**
PMB endometrial thickness	Sample if **>4 mm** (some units use 5 mm)
Shock index	**≥0.9** = not an OPD patient

Exam traps

- **"Cervical erosion" treated with cautery.** Wrong: an ectropion is a physiological variant; treat only if symptomatic **after** normal screening and treated infection. Never ablate in pregnancy.
- **PCB treated with antibiotics and reassurance.** Wrong: PCB needs a speculum look, a screening test, and biopsy of anything abnormal; persistent PCB with a normal screen needs **colposcopy**.
- **Candida or BV listed as causes of cervicitis.** Wrong: they are **vaginal** infections; mucopus from the os means **gonorrhoea/chlamydia** (columnar-cell bugs).
- **Doxycycline to a woman who "might be pregnant".** Wrong: do the pregnancy test first; in pregnancy use **azithromycin 1 g stat**.
- **Polyp avulsed and thrown away.** Wrong: **every** polyp goes for histology; in PMB also assess the endometrium.
- **Repeating the same syndromic packet after failure.** Wrong: check adherence and partner treatment, then **swab for NAAT/culture and sensitivities** and think of other diagnoses (cancer, TB, schistosomiasis).
- **Speculum for bleeding at 32 weeks.** Wrong: exclude **placenta praevia by ultrasound** first.
- **Ablating a lesion that looks invasive** "because VIA was positive". Wrong: a growth or ulcer is **suspicious for cancer** → biopsy and refer.

Questions seniors ask

Viva questions

1. *What is the transformation zone and why does it matter?* The area between the original and the new SCJ where columnar epithelium undergoes squamous metaplasia; metaplastic cells are vulnerable to HPV, so

almost all CIN and squamous cancers start here; it is what screening must sample.

2. *Why is "cervical erosion" a bad term?* The epithelium is intact; it is columnar epithelium on the ectocervix (ectopy), a physiological change.
3. *What organisms cause mucopurulent cervicitis? C. trachomatis* and *N. gonorrhoeae* mainly; also *M. genitalium*, trichomonas, HSV.
4. *Why doesn't candida cause cervicitis?* Candida and BV are vaginal infections of the squamous-lined vagina; GC/CT infect columnar endocervical cells.
5. *How do you remove a cervical polyp and what must you always do?* Avulsion by twisting with polyp forceps, haemostasis, and **always histology**.
6. *What is the significance of Nabothian cysts?* Benign; they mark the TZ; no treatment.
7. *Why are cervical fibroids surgically dangerous?* They distort the ureters and bladder and bleed; ureteric injury.
8. *Give the indications for history- and ultrasound-indicated cerclage.* ≥3 mid-trimester losses/PTB → 12–14 weeks; previous loss/PTB + cervix ≤25 mm before 24 weeks.
9. *A woman fails syndromic treatment: what next?* Check adherence and partner treatment; re-examine; swab for culture/NAAT and sensitivities; consider resistance, other diagnoses (cancer, TB, schistosomiasis), HIV.

On the ward round

- **Where exactly is the discharge coming from, and why does it matter?** From the os = cervical infection (treat GC + CT); coating the vaginal walls = vaginitis (BV/TV/candida). That single observation chooses the drugs.
- **What makes the risk assessment positive?** Partner with urethral discharge or genital ulcer, age under about 21–25, a new partner in the last 3 months, or more than one partner. Any one → treat for cervical infection even if the cervix looks normal.
- **How do you know this red area is an ectropion and not cancer?** Flat, symmetrical around the os, regular border, grape-like columnar villi, no dense acetowhitening, SCJ seen; cancer is raised or ulcerated, irregular, hard and friable. If in doubt, biopsy.
- **This woman is HPV-positive and VIA-positive: what now?** If eligible (whole lesion and SCJ seen, <75%, no canal extension, no suspicion of cancer, not pregnant) → thermal ablation or cryotherapy the same day; if not eligible → LEEP; if suspicious → biopsy. Re-test at 12 months.
- **Why do you send a benign-looking polyp for histology?** Rarely it contains dysplasia or carcinoma, and a "polyp" may be a prolapsed fibroid or a polypoid cancer; histology is cheap insurance.
- **When would you not avulse a polyp in OPD?** Broad (sessile) base, large or very vascular, stalk high in the canal or from the cavity (prolapsed fibroid), or suspicious appearance → theatre (EUA/hysteroscopy).
- **She has pyometra at 70 years: what are you worried about?** Cervical or endometrial cancer causing stenosis: drain, give antibiotics, and biopsy.
- **Why is a cervical fibroid dangerous at hysterectomy?** It displaces the ureters (1.5–2 cm lateral to the cervix, under the uterine artery) and bladder: ureteric injury.

Mnemonics

- **"Red cervix: ECCC"**: **E**ctropion, **C**ervicitis, **C**IN/Cancer, **C**ontraceptive pill (hormonal ectopy); then look for the border and the discharge.
- **Causes of PCB: "PEACE"**: **P**olyp, **E**ctropion, **A**trophic vaginitis, **C**ervicitis, **E**ndometrial/cervical cancer (cancer must be excluded first!).
- **The 4 Cs of STI care**: **C**ompliance, **C**ounselling, **C**ondoms, **C**ontact tracing (some add a 5th C: **C**linical follow-up).
- **"Water under the bridge"**: the ureter passes under the uterine artery 1.5–2 cm lateral to the cervix: danger zone for cervical fibroids and cancer surgery.

- **Columnar-loving bugs = GC and CT** (the "Cervix Club"); **squamous/vaginal bugs = BV, TV, Candida**.
- **Polyp rule: "Twist, dab, pot"**: twist it off, dab the base (silver nitrate), put it in the pot for histology.

OSCE tips

> EXAM FAVOURITE
> **OSCE: speculum and VIA station:** introduce, consent, chaperone, empty bladder, lithotomy, inspect vulva, warm and lubricate speculum, insert obliquely and rotate, open to visualise the cervix, **describe the SCJ, discharge and lesions**, apply 3–5% acetic acid, wait 1 minute, **report VIA negative / positive / suspicious for cancer**, close and withdraw the speculum carefully, explain findings and the plan.

- **Polypectomy station**: consent, exclude pregnancy, identify the stalk, grasp at the base, twist in one direction, haemostasis, specimen in formalin for histology, state when you would send her to theatre instead.
- **Counselling station (ectropion)**: explain that it is a normal variant caused by hormones, not a wound or cancer; screening confirms this; treatment only if troublesome.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define the transformation zone (TZ) of the cervix.	The area between the **original** and the **new (functional) squamocolumnar junction**, where columnar epithelium is being replaced by squamous epithelium (**squamous metaplasia**). Almost all CIN and squamous cancers arise here, so screening must sample it.
2	What epithelium lines (a) the ectocervix and (b) the endocervix?	(a) **Non-keratinised stratified squamous** epithelium. (b) A **single layer of mucus-secreting columnar** epithelium arranged in crypts.
3	What is a cervical ectropion and why does it happen?	Endocervical **columnar epithelium visible on the ectocervix** (red, velvety area around the os). Caused by **oestrogen** eversion of the cervix — puberty, **pregnancy**, **combined pill**. A physiological variant, not a disease.
4	Why does an ectropion cause post-coital bleeding and discharge?	Columnar epithelium is **one cell thick** over fragile capillaries (bleeds on contact) and is **mucus-secreting** (more mucoid discharge).
5	When should an ectropion be treated, and how?	Only if **symptomatic** (troublesome discharge/PCB) **after** a normal screening test and treated infection. Options: switch from COC to progestogen-only method; **ablation** (cryotherapy, thermal ablation, diathermy). Never in pregnancy.
6	How does a Nabothian cyst form and what is its clinical significance?	Squamous metaplasia covers the mouth of an endocervical crypt, trapping mucus. It is **benign**, needs **no treatment**, and **marks the transformation zone**.
7	Which organisms cause mucopurulent cervicitis and why the endocervix?	**Chlamydia trachomatis** and **Neisseria gonorrhoeae** (also Mycoplasma genitalium, HSV, trichomonas). GC and CT infect **columnar** cells, so they target the endocervix and ectropion.
8	Uganda syndromic treatment of cervical infection (non-pregnant).	**Cefixime 400 mg PO stat** (or **ceftriaxone 1 g IM stat**) **+ doxycycline 100 mg PO 12-hourly for 7 days**, usually with **metronidazole 2 g PO stat** for vaginal infection. Treat partner(s), condoms, HIV test, review in 7 days.
9	What replaces doxycycline for chlamydia cover in pregnancy?	**Azithromycin 1 g PO single dose** (or erythromycin 500 mg PO 6-hourly for 7 days). Doxycycline is contraindicated in pregnancy and breastfeeding.
10	What makes the risk assessment for cervical infection positive in syndromic management?	Any of: **partner with urethral discharge/genital ulcer**, **young age** (under about 21–25), **new partner in the last 3 months**, or **more than one partner**. Positive risk assessment = treat for cervical infection even if the cervix looks normal.
11	What are the 4 Cs of STI management?	**Compliance**, **Counselling**, **Condoms**, **Contact tracing** (partner notification and treatment).
12	Outpatient treatment of mild–moderate PID.	**Ceftriaxone 1 g IM stat** + **doxycycline 100 mg PO 12-hourly for 14 days** + **metronidazole 400–500 mg PO 12-hourly for 14 days**. Pregnancy test first; admit if severe, pregnant, TOA or no response in 72 h.
13	How is a cervical polyp removed in clinic?	Grasp the base with **sponge-holding (polyp) forceps**, **twist (avulse)** until it detaches, achieve haemostasis (pressure/**silver nitrate**), and **always send for histology**. Refer to theatre if broad-based, large, very vascular or arising high in the canal.

#	Question	Answer
14	Why assess the endometrium when a post-menopausal woman has a cervical polyp?	Coexisting **endometrial pathology** (polyps, hyperplasia, cancer) is common and the bleeding may not be from the cervical polyp. Do **TVUS endometrial thickness ± endometrial biopsy**.
15	What is the main surgical danger of a cervical fibroid?	It distorts the pelvic anatomy and displaces the **ureters** (and bladder), which lie about 1.5–2 cm lateral to the cervix under the uterine artery — risk of **ureteric injury** and haemorrhage at myomectomy/hysterectomy.
16	Pyometra in an elderly woman — what must you exclude?	**Cervical or endometrial cancer** (causing stenosis). Drain (dilate the cervix), give antibiotics and take **biopsies**.
17	Indications for history- and ultrasound-indicated cerclage (RCOG).	**History-indicated** at **12–14 weeks**: ≥3 previous mid-trimester losses/preterm births. **Ultrasound-indicated**: previous mid-trimester loss/PTB **and** transvaginal cervical length **≤25 mm before 24 weeks**. Remove at **36–37 weeks**.
18	How do you tell an ectropion from early cervical cancer by eye?	Ectropion: **flat, symmetrical around the os, regular border**, velvety/grape-like, no dense acetowhitening. Cancer: **raised or ulcerated, irregular, asymmetric, hard, friable/necrotic**. If unsure — **biopsy**.
19	What is a positive VIA test?	After applying **3–5% acetic acid** and waiting **1 minute**: a **dense, well-defined acetowhite lesion touching or near the SCJ**. A cauliflower growth or ulcer = **suspicious for cancer** (biopsy/refer).
20	A woman fails a full course of syndromic treatment for cervicitis. Next steps?	Check **adherence** and whether the **partner was treated**; re-examine; take an **endocervical swab for NAAT/culture and sensitivities** (AMR to doxycycline, azithromycin and cefixime is documented in Uganda); consider PID, HIV, **cancer**, TB/schistosomiasis.
21	First 5 minutes: which findings turn a woman with 'cervical bleeding or discharge' from an OPD patient into an emergency?	**Shock index ≥0.9** or signs of shock, **heavy active bleeding**, a **positive pregnancy test with pain/bleeding** (ectopic until proven otherwise), **fever with peritonism** (severe PID/TOA), and **any bleeding after about 20 weeks** before placenta praevia has been excluded by ultrasound (no speculum until then).
22	In what order should you describe a speculum examination of the cervix?	(a) **Discharge**: amount, colour, and whether it comes **from the os** or coats the **vaginal walls**; (b) the **SCJ**: seen or inside the canal; (c) the ectocervix: ectropion, Nabothian cysts, **friability**; (d) any **mass**: polyp (stalk?), fibroid, growth, ulcer; (e) vaginal walls. Take swabs/HPV sample **before** acetic acid, then VIA.
23	State the Amsel criteria for bacterial vaginosis.	**3 of 4**: thin grey homogeneous discharge; **vaginal pH >4.5**; **positive whiff test** (fishy amine smell with 10% KOH); **clue cells** on wet mount.
24	A 29-year-old HIV-positive woman is HPV-positive; VIA shows a dense acetowhite lesion at the SCJ covering about 25% of the ectocervix, SCJ fully seen, no canal extension, not pregnant. Next step?	She is **triage (VIA) positive and eligible for ablation → thermal ablation the same day** (≥100 °C, 20–30 s per application) or cryotherapy (3-5-3). Advise watery discharge 2–4 weeks, no intercourse/tampons for 4 weeks; **re-test at 12 months**. A growth or ulcer would instead need **biopsy and referral**.
25	Post-menopausal bleeding with a cervical polyp: after avulsion, what endometrial thickness on TVUS triggers endometrial sampling?	**>4 mm** in most units (some use 5 mm). Sample with Pipelle or D&C/hysteroscopy, because coexisting endometrial pathology is common and the polyp may not be the source.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 19-year-old university student on the combined oral contraceptive pill attends a clinic in Kampala with occasional post-coital spotting. She is HIV-negative and has one partner of 2 years with no symptoms. Speculum shows a symmetrical, red, velvety area around the external os with a regular border; there is clear mucoid discharge and no mucopus. Which is the MOST likely diagnosis?

A Acute gonococcal cervicitis
B Cervical intraepithelial neoplasia grade 3
C Cervical ectropion
D Endocervical polyp
E Trichomonas vaginalis infection

2. A 24-year-old P1 attends HC IV with yellow vaginal discharge for 10 days. Her husband has had painful micturition with a urethral discharge. She is not pregnant (negative urine hCG) and has no abdominal pain. Speculum shows mucopus from an oedematous, friable cervix; there is no cervical motion tenderness. No laboratory tests are available. What is the MOST appropriate treatment for her?

A Ceftriaxone 1 g IM stat (or cefixime 400 mg PO stat) + doxycycline 100 mg PO 12-hourly for 7 days + metronidazole 2 g PO stat, and treat the husband
B Metronidazole 2 g PO stat and clotrimazole 500 mg pessary stat
C Amoxicillin 500 mg PO 8-hourly for 5 days
D Ceftriaxone 1 g IM stat + doxycycline 100 mg 12-hourly + metronidazole 400 mg 12-hourly for 14 days, and admit
E Refer for culture and give no treatment until results are available

3. A 27-year-old woman who is 14 weeks pregnant has mucopurulent cervical discharge and her partner has urethral discharge. Which regimen is MOST appropriate?

A Ceftriaxone 1 g IM stat + doxycycline 100 mg 12-hourly for 7 days
B Ciprofloxacin 500 mg stat + doxycycline 100 mg 12-hourly for 7 days
C Fluconazole 150 mg stat + metronidazole 2 g stat
D Doxycycline 100 mg 12-hourly for 14 days alone
E Ceftriaxone 1 g IM stat (or cefixime 400 mg PO stat) + azithromycin 1 g PO stat

4. A 45-year-old P5, never screened, presents to Mengo gynaecology clinic with 4 months of post-coital bleeding. On speculum there is a 3 cm irregular, friable, exophytic growth on the anterior lip of the cervix that bleeds heavily on touch. What is the MOST appropriate next step?

A Treat syndromically for cervicitis and review in 2 weeks
B Take a punch biopsy of the lesion for histology and refer to gynae-oncology
C Perform VIA and, if positive, offer thermal ablation today
D Avulse the growth with polyp forceps
E Reassure her that this is an ectropion

5. A 48-year-old P6 has inter-menstrual spotting. Speculum shows a 1 cm smooth, red, pedunculated mass protruding from the external os; the stalk is thin and visible; VIA of the surrounding cervix is negative. What is the BEST management?

A Reassure and discharge; polyps never become malignant
B Hysterectomy
C Cryotherapy of the mass
D Avulsion with sponge-holding forceps, haemostasis and histology of the specimen
E Oral tranexamic acid and review in 3 months

6. A 62-year-old woman presents with 3 weeks of offensive vaginal discharge and low-grade fever. She has been menopausal for 12 years. Ultrasound shows a fluid-filled uterine cavity; the cervical os is pinpoint and cannot be probed in clinic. Which statement about her management is MOST accurate? Advanced

A She needs antibiotics only; the fluid will drain spontaneously
B Hysterectomy should be done immediately without prior drainage
C Cervical dilatation to drain the pyometra, antibiotics, and biopsy of the cervix and endometrium to exclude malignancy
D Insert vaginal clotrimazole pessaries for 6 nights
E This is physiological post-menopausal cervical stenosis; reassure

7. A 30-year-old woman with vaginal discharge is found to have lower abdominal pain, temperature 38.2 °C, cervical motion tenderness and bilateral adnexal tenderness. Urine hCG is negative. She is eating and drinking, and there is no peritonism or mass on ultrasound. What is the MOST appropriate treatment?

A Ceftriaxone 1 g IM stat + doxycycline 100 mg PO 12-hourly for 14 days + metronidazole 400–500 mg PO 12-hourly for 14 days
B Cefixime 400 mg stat + doxycycline 100 mg 12-hourly for 7 days
C Metronidazole 2 g PO stat
D Laparotomy
E Azithromycin 1 g stat only

8. During routine VIA screening of a 34-year-old woman, you notice several smooth, yellow-white, translucent, dome-shaped blebs on the ectocervix near the os. VIA is negative and she has no symptoms. What should you do?

A Refer for LEEP
B Take a biopsy of each cyst
C Treat with cryotherapy
D Treat for chlamydia
E Reassure her; these are Nabothian cysts and need no treatment

9. A 28-year-old woman has had two courses of syndromic treatment for mucopurulent cervicitis (cefixime, doxycycline, metronidazole) at a HC III. She reports taking all the tablets. Her partner has not been treated. She still has mucopus. What is the MOST important next step? Advanced

A Give a third identical course
B Ensure partner treatment and take an endocervical swab for NAAT/culture and sensitivity
C Perform a hysterectomy
D Give fluconazole 150 mg stat
E Reassure her that chronic cervicitis is normal

10. Which of the following BEST explains why Chlamydia trachomatis and Neisseria gonorrhoeae infect the endocervix rather than the vaginal walls?

- A They require the alkaline pH of the vagina
- B They are killed by cervical mucus
- C They infect keratinised epithelium
- D They preferentially infect columnar epithelial cells
- E They only infect post-menopausal women

11. A 36-year-old P3 at 13 weeks' gestation had three previous painless mid-trimester losses at 18–20 weeks, each with little pain before the membranes bulged. There is no infection and the ultrasound is normal. What is the MOST appropriate management? Advanced

- A History-indicated (McDonald) cerclage now, with removal at 36–37 weeks
- B Bed rest until term
- C Emergency cerclage only if she presents with bulging membranes
- D Hysterectomy after delivery
- E Tocolysis with nifedipine from now until 34 weeks

12. A 42-year-old woman with a 12-week-size pelvic mass and urinary frequency is found on ultrasound to have a 9 cm fibroid arising from the supravaginal cervix. She has completed her family. What is the MOST important surgical consideration when she undergoes hysterectomy at a regional referral hospital? Advanced

- A The ovaries must always be removed
- B Spinal anaesthesia is contraindicated
- C The ureters may be displaced by the fibroid and must be identified to avoid injury
- D The procedure can be done safely at HC III
- E A vaginal hysterectomy is always preferred

13. A 32-year-old woman at 34 weeks' gestation presents with painless vaginal spotting after intercourse. The fetal heart is normal. She has not had an ultrasound this pregnancy. What is the MOST appropriate FIRST step before any speculum or digital examination?

- A Digital vaginal examination to assess the cervix
- B Speculum examination and VIA
- C Immediate caesarean section
- D Syndromic STI treatment
- E Ultrasound to locate the placenta

14. A 22-year-old woman has a 1-week history of curdy white vaginal discharge with intense vulval itching. Her partner is asymptomatic, she has one partner, and the risk assessment is negative. Speculum shows white plaques on the vaginal walls; the cervix is healthy with no mucopus. She is not pregnant. What is the BEST treatment?

- A Ceftriaxone 1 g IM stat + doxycycline 7 days
- B Clotrimazole 500 mg vaginal pessary stat (or fluconazole 150 mg PO stat)
- C Metronidazole 2 g PO stat only
- D Doxycycline 100 mg 12-hourly for 14 days
- E Cryotherapy of the cervix

15. A 26-year-old woman has troublesome mucoid discharge and post-coital spotting. HPV test is negative, NAAT for chlamydia/gonorrhoea is negative, and speculum shows a large ectropion. She has been on the combined pill for 3 years and wants the symptoms to stop. What is the MOST appropriate next step?

A Radical trachelectomy
B Long-term doxycycline
C Cone biopsy
D Discuss switching to a progestogen-only method and, if symptoms persist, offer ablation (cryotherapy/thermal ablation)
E Tell her ectropion is premalignant and requires LEEP

Answers & explanations

1. C · Cervical ectropion

C is correct. A symmetrical, regular-bordered, red velvety area around the os in a young woman on the COC is an **ectropion** — oestrogen-driven eversion of columnar epithelium, which bleeds on contact. **A** is wrong: gonococcal cervicitis gives mucopurulent discharge from the os and an oedematous, friable cervix, and she has no risk factors. **B** is wrong: CIN is usually invisible without acetic acid and does not cause a regular symmetrical red area. **D** is wrong: a polyp is a discrete pedunculated mass protruding from the os. **E** is wrong: trichomonas causes frothy yellow-green discharge, itch and a punctate strawberry cervix.

2. A · Ceftriaxone 1 g IM stat (or cefixime 400 mg PO stat) + doxycycline 100 mg PO 12-hourly for 7 days + metronidazole 2 g PO stat, and treat the husband

A is correct. Mucopus + friable cervix + partner with urethral discharge = **cervical infection** (gonorrhoea/chlamydia). Syndromic treatment covers GC (ceftriaxone/cefixime), CT (doxycycline) and vaginal pathogens (metronidazole), with **partner treatment**. **B** is wrong: it treats only vaginal infection (BV/TV/candida) and leaves GC/CT untreated. **C** is wrong: amoxicillin is not a recommended STI regimen. **D** is wrong: this is a 14-day PID regimen and she has no pain or cervical motion tenderness; admission is not needed. **E** is wrong: syndromic management exists precisely so treatment is not delayed where labs are unavailable.

3. E · Ceftriaxone 1 g IM stat (or cefixime 400 mg PO stat) + azithromycin 1 g PO stat

E is correct. In pregnancy, gonorrhoea is covered with a cephalosporin and chlamydia with **azithromycin 1 g stat** (or erythromycin) because **doxycycline is contraindicated**. **A** and **D** are wrong because they use doxycycline in pregnancy (and D omits gonorrhoea cover). **B** is wrong: ciprofloxacin has high gonococcal resistance, is avoided in pregnancy, and it again uses doxycycline. **C** is wrong: fluconazole is avoided in pregnancy and neither drug covers GC/CT.

4. B · Take a punch biopsy of the lesion for histology and refer to gynae-oncology

B is correct. An irregular, friable, exophytic growth is **suspicious for invasive cancer**; the diagnosis requires **histology**, and she needs staging and referral. **A** is wrong: treating as infection delays a cancer diagnosis. **C** is wrong: VIA is a screening test for women **without** a visible lesion, and ablation is **contraindicated** when cancer is suspected. **D** is wrong: this is not a pedunculated polyp; avulsion would cause haemorrhage and a poor specimen. **E** is wrong: ectropion is flat, regular and symmetrical.

5. D · Avulsion with sponge-holding forceps, haemostasis and histology of the specimen

D is correct. A small, thin-stalked endocervical polyp is removed in clinic by **twisting (avulsion)**, haemostasis with pressure/silver nitrate, and **always histology**. **A** is wrong: malignancy is rare but real, and symptoms warrant removal. **B** is wrong: hysterectomy is unnecessarily radical. **C** is wrong: ablation destroys the tissue so no histology is obtained. **E** is wrong: it treats the symptom and leaves the lesion and its histology unknown.

6. C · Cervical dilatation to drain the pyometra, antibiotics, and biopsy of the cervix and endometrium to exclude malignancy

C is correct. Pyometra results from **cervical stenosis**; drainage (dilatation) + antibiotics treat it, and in an elderly woman **cervical or endometrial cancer must be excluded** by biopsy. **A** is wrong: a stenosed canal will not drain spontaneously. **B** is wrong: operating on an infected, undiagnosed uterus is dangerous; drain, diagnose, then plan. **D** is wrong: this is not candidiasis. **E** is wrong: stenosis may be common after menopause, but pyometra with fever is not physiological.

7. A · Ceftriaxone 1 g IM stat + doxycycline 100 mg PO 12-hourly for 14 days + metronidazole 400–500 mg PO 12-hourly for 14 days

A is correct. She has **PID** (cervicitis that has ascended). Mild–moderate PID in a woman tolerating oral drugs is treated as an outpatient with ceftriaxone + **14 days** of doxycycline and metronidazole (anaerobic cover), with review in 72 hours. **B** is wrong: 7 days is the uncomplicated cervicitis regimen and lacks anaerobic cover. **C** is wrong: it misses GC and CT. **D** is wrong: surgery is for a ruptured/unresponsive TOA or diagnostic doubt. **E** is wrong: single-dose azithromycin is inadequate for PID.

8. E · Reassure her; these are Nabothian cysts and need no treatment

E is correct. These are **Nabothian cysts** — benign mucus-retention cysts formed by squamous metaplasia covering crypt openings in the TZ. No treatment is needed. **A**, **B** and **C** are wrong: they are unnecessary interventions for a benign finding in a VIA-negative woman. **D** is wrong: Nabothian cysts are not a sign of infection.

9. B · Ensure partner treatment and take an endocervical swab for NAAT/culture and sensitivity

B is correct. The commonest reasons for failure are **re-infection from an untreated partner** and **antimicrobial resistance** (high resistance to doxycycline, azithromycin and cefixime has been reported in Uganda). Treat the partner and obtain a diagnosis with sensitivities. **A** is wrong: repeating the same drugs ignores both causes. **C** is wrong: grossly inappropriate. **D** is wrong: candida does not cause mucopus. **E** is wrong: persistent mucopus needs investigation (and screening for cancer).

10. D · They preferentially infect columnar epithelial cells

D is correct. GC and CT are tropic for **columnar** epithelium, which lines the endocervix (and ectropion). This is why ectopy increases susceptibility and why they cause mucopurulent cervicitis. **A** is wrong: the normal vaginal pH is acidic (about 4). **B** is wrong: mucus does not protect against them. **C** is wrong: the vagina is non-keratinised squamous, and the organisms do not favour squamous epithelium. **E** is wrong: they infect sexually active women of any age, most commonly the young.

11. A · History-indicated (McDonald) cerclage now, with removal at 36–37 weeks

A is correct. Three or more previous mid-trimester losses/preterm births is the RCOG indication for **history-indicated cerclage at 12–14 weeks**; it is removed at 36–37 weeks. **B** is wrong: bed rest has no proven benefit and causes thrombosis risk. **C** is wrong: waiting for bulging membranes gives a much poorer outcome. **D** is irrelevant. **E** is wrong: tocolysis does not treat a structurally weak cervix and prolonged use is not recommended.

12. C · The ureters may be displaced by the fibroid and must be identified to avoid injury

C is correct. Cervical fibroids distort pelvic anatomy and displace the **ureters**, which normally run 1.5–2 cm lateral to the cervix under the uterine artery; this is a classic cause of ureteric injury. **A** is wrong: ovaries are conserved in a 42-year-old without ovarian pathology. **B** is wrong: regional anaesthesia is not contraindicated. **D** is wrong: this is complex surgery needing an experienced surgeon at a referral hospital. **E** is wrong: a large cervical fibroid usually makes vaginal hysterectomy difficult.

13. E · Ultrasound to locate the placenta

E is correct. Any bleeding in late pregnancy is **placenta praevia until proven otherwise**; the placenta must be localised before vaginal examination. Once praevia is excluded, a gentle speculum can look for ectropion, polyp, cervicitis or cancer. **A** is wrong and dangerous: a digital examination can provoke massive haemorrhage in praevia. **B** is wrong as the first step, and VIA is not done in pregnancy for routine screening. **C** is wrong: there is no indication. **D** is wrong: it treats a presumed cause without assessment.

14. B · Clotrimazole 500 mg vaginal pessary stat (or fluconazole 150 mg PO stat)

B is correct. Curdy discharge, itch, vulvovaginal erythema and a healthy cervix with negative risk assessment = **vulvovaginal candidiasis**, treated with clotrimazole or fluconazole (not in pregnancy). **A** and **D** are wrong: there is no evidence of cervical infection. **C** is wrong: metronidazole treats BV/TV, not candida (in syndromic practice it may be added, but on its own it misses the diagnosis). **E** is wrong and irrelevant.

15. D · Discuss switching to a progestogen-only method and, if symptoms persist, offer ablation (cryotherapy/thermal ablation)

D is correct. Once cancer and infection are excluded, a **symptomatic** ectropion may be managed by reducing oestrogen exposure (stopping COC) and, if needed, **ablation**. **A** and **C** are wrong: excisional/radical procedures are for neoplasia. **B** is wrong: there is no infection and antibiotics will not change the epithelium. **E** is wrong: ectropion is physiological, not premalignant.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Discharge and a worried husband

Nakato, a 23-year-old P1+0 hairdresser, comes to a HC IV in Wakiso with a 2-week history of yellow vaginal discharge and spotting after sex. She delivered 8 months ago and is using DMPA. Her husband recently travelled and now has pain on passing urine.

1. What focused history do you take, and why?

> Character of discharge (mucopurulent vs curdy vs frothy vs fishy) to separate cervical from vaginal causes; **lower abdominal pain, fever, dyspareunia** (PID); **risk assessment** — partner symptoms (positive here), new/multiple partners; **LMP and pregnancy possibility** (DMPA, breastfeeding — affects doxycycline); breastfeeding status; HIV status and last test; cervical screening history; previous STI treatment.

2. Examination shows T 36.8 °C, soft non-tender abdomen. Speculum: mucopus from a red, oedematous cervix that bleeds on touching with a swab. No cervical motion or adnexal tenderness. What is your diagnosis and differential?

> **Acute mucopurulent cervicitis**, most likely gonococcal and/or chlamydial (partner with urethritis = positive risk assessment). Differentials: ectropion with superadded infection, trichomonas (strawberry cervix, frothy discharge), early PID (no — no CMT/adnexal tenderness), and — because she has contact bleeding — CIN/cancer to be excluded once inflammation settles.

3. What investigations are useful at a HC IV?

> **Urine pregnancy test**; HIV test and syphilis rapid test; endocervical swab for **GeneXpert CT/NG** or gram stain if available (not required to start treatment); VIA **after** treatment (inflammation causes false positives) — or an HPV sample.

4. She is not pregnant but is breastfeeding. Write the treatment plan.

> Syndromic treatment of cervical + vaginal infection: **ceftriaxone 1 g IM stat (or cefixime 400 mg PO stat)** + chlamydia cover — as she is breastfeeding, **azithromycin 1 g PO stat** is preferred over doxycycline — + **metronidazole 2 g PO stat** (or 400–500 mg 12-hourly for 7 days to minimise milk levels; brief interruption of breastfeeding after a 2 g dose is sometimes advised). **4 Cs**: compliance, counselling, **condoms**, **contact slip** — the husband is treated with the urethral discharge regimen. Abstain until 7 days after both complete treatment. Review in 7 days.

5. At review she is better but still has some post-coital spotting. What now?

> Re-examine: if cervicitis has resolved, do **VIA/HPV testing**; if an ectropion is present and screening is negative — reassure (DMPA/postpartum). If a lesion is seen or bleeding persists → colposcopy/biopsy at a referral hospital. If mucopus persists → check partner treatment and send swab for culture/sensitivity (AMR).

Take-away: Mucopus + friable cervix + symptomatic partner = cervicitis: treat GC and CT syndromically (adjusted for pregnancy/breastfeeding), treat the partner, and screen the cervix once inflammation settles — contact bleeding is never simply 'infection' until the cervix has been screened.

Case 2: Post-coital bleeding at 45

Mrs Akello, a 45-year-old P6+0 teacher from Lira, is referred to Mengo Hospital gynaecology clinic with 3 months of post-coital bleeding and occasional inter-menstrual spotting. She has never been screened. She is HIV-positive on TLD with a suppressed viral load.

1. Why is this presentation worrying and what history do you add?

> PCB in a **45-year-old, never-screened, HIV-positive** woman — cervical cancer must be excluded first (HIV raises the risk several-fold). Ask: offensive/blood-stained discharge, weight loss, back/leg pain, leg swelling, urinary/faecal leakage (advanced disease), menstrual pattern, contraception, partner symptoms, CD4 history, previous STIs.

2. Speculum shows a smooth, 1.5 cm, red, pedunculated mass protruding from the os with a thin stalk. The rest of the ectocervix looks healthy. What is your impression and what do you do next?

> **Endocervical polyp** (differentials: prolapsed submucous fibroid, polypoid carcinoma, ectropion). Do **VIA/HPV testing** of the rest of the cervix, then **avulse** the polyp with sponge-holding forceps (twist), secure haemostasis with pressure/silver nitrate, and put it in **formalin for histology**.

3. VIA is positive: a dense acetowhite area at 12 o'clock touching the SCJ, occupying under 50% of the cervix, fully visible, no features of cancer. What is the plan?

> She is eligible for **ablation**: offer **thermal ablation** (or cryotherapy) in the same visit (screen-and-treat), with counselling. For a woman living with HIV WHO recommends screen–triage–treat where possible (HPV test then VIA triage); follow-up **HPV/VIA at 12 months** after treatment. Had the lesion been large (>75%), extended into the canal or looked suspicious → LEEP or biopsy at a referral centre.

4. Histology returns: benign endocervical polyp. What further follow-up does she need?

> Review the bleeding: if it has stopped, routine care. She needs **post-treatment screening at 1 year**, then regular screening as a woman living with HIV (WHO: every 3–5 years with HPV, or per Uganda programme). Continue ART adherence counselling. If bleeding persists → colposcopy and endometrial assessment.

5. How would you present her on the ward round?

> 'Mrs A, 45, P6, HIV-positive virally suppressed, never screened, with 3 months of PCB. Speculum showed a 1.5 cm pedunculated endocervical polyp which was avulsed and sent for histology — benign. VIA was positive with an ablation-eligible lesion, treated with thermal ablation. Plan: repeat HPV/VIA in 12 months, continue ART, return if bleeding recurs.'

Take-away: Treating the obvious benign lesion (polyp) is only half the job: the whole cervix must be screened, the polyp must go for histology, and women living with HIV need screen-and-treat plus 12-month follow-up.

Case 3: The mass that wasn't a polyp

Mrs Namutebi, a 39-year-old P2, attends a regional referral hospital with 6 months of heavy periods, a 'lump' felt at the vaginal opening in the last week, and cramping lower abdominal pain. Hb is 7.2 g/dL.

1. What are your differentials for a mass at the os with heavy bleeding?

> **Prolapsed submucous fibroid (fibroid polyp)**, large endocervical polyp, cervical fibroid, polypoid cervical or endometrial cancer/sarcoma, uterine inversion (post-partum — not here), products of conception (check hCG).

2. On speculum a 5 cm firm, pale, smooth mass fills the vagina; its stalk passes through a dilated os into the uterine cavity. What investigations do you do?

> **Pregnancy test**; **FBC** (Hb 7.2), group and cross-match; **pelvic ultrasound** to confirm the stalk arises from the uterine cavity and to look for other fibroids; renal function if retention; VIA/HPV (if the cervix can be seen) for screening.

3. What is the management?

> **Correct anaemia**: transfuse if symptomatic/very low before surgery, then **ferrous sulphate 200 mg PO 8-hourly** (with folic acid). **Vaginal myomectomy in theatre** under spinal/GA: grasp the fibroid, clamp and ligate the pedicle (or twist off), achieve haemostasis, **histology**. Antibiotics if the mass is necrotic/infected. Do not attempt avulsion in OPD — heavy bleeding from a thick stalk.

4. What complications should you anticipate and what follow-up is required?

> Haemorrhage from the pedicle; **uterine inversion** if traction is excessive; infection; perforation if the base is high; recurrence from other fibroids. Follow-up with histology (exclude leiomyosarcoma), Hb check, and ultrasound for residual fibroids; discuss further fertility or definitive options.

> **Take-away:** A large, firm, pale 'polyp' with heavy bleeding is often a prolapsed submucous fibroid: image it, correct anaemia, and remove it in theatre with histology — never pull on it in clinic.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.